

Status note for arXiv:1005.3634, Problem 1

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Status

This is a lightweight literature-status packet. The extracted problem is *already answered in later literature*; no new proof is claimed here.

Source question

The source paper is F. Bermudez, A. Bonilla, F. Martinez-Gimenez, and A. Peris, *Li-Yorke and distributionally chaotic operators*, J. Math. Anal. Appl. 373 (2011), 83–93, DOI 10.1016/j.jmaa.2010.06.011, arXiv:1005.3634.

In Section 4, “Distributionally chaotic operators”, immediately after Proposition 16, the paper asks:

Does every distributionally chaotic operator $T : X \rightarrow X$ admit a distributionally irregular vector?

In the arXiv source this is Problem 1.

Answering theorem

The answer source is N. C. Bernardes Jr., A. Bonilla, V. Muller, and A. Peris, *Distributional chaos for linear operators*, J. Funct. Anal. 265 (9) (2013), 2143–2163, DOI 10.1016/j.jfa.2013.06.019.

The relevant result is Theorem 12 of the 2013 JFA paper. Later accessible sources restate it in the following form: for a bounded linear operator T on a Banach space X , the following conditions are equivalent: T satisfies the Distributional Chaos Criterion, T has a distributionally irregular vector, T is distributionally chaotic, and T admits a distributionally chaotic pair. Thus the implication

$$T \text{ distributionally chaotic} \implies T \text{ has a distributionally irregular vector}$$

is exactly the affirmative answer to Problem 1.

Provenance check

Crossref metadata for the 2013 JFA article records a reference to the 2011 source article by DOI 10.1016/j.jmaa.2010.06.011, title, journal, volume, issue, first page, and year. This supports placement in `literature_already_answered/`: the later answer source is separate from, and bibliographically connected to, the source problem paper.

The packet includes the source arXiv PDF and Crossref metadata for both DOIs. The decisive 2013 JFA paper does not appear as a local arXiv PDF in the corpus; therefore the packet also includes the accessible witness arXiv:1807.05191, which restates Bernardes–Bonilla–Muller–Peris Theorem 12 as Theorem 1.1.

Conclusion

The target arXiv:1005.3634 is not viable for a new full proof or counterexample on this question. Problem 1 is already solved in the affirmative by Bernardes–Bonilla–Muller–Peris, Theorem 12.

References

- [1] F. Bermudez, A. Bonilla, F. Martinez-Gimenez, and A. Peris, *Li-Yorke and distributionally chaotic operators*, J. Math. Anal. Appl. 373 (2011), 83–93.
- [2] N. C. Bernardes Jr., A. Bonilla, V. Muller, and A. Peris, *Distributional chaos for linear operators*, J. Funct. Anal. 265 (9) (2013), 2143–2163.
- [3] C. Chen, *Distributional chaos for weighted translation operators on groups*, arXiv:1807.05191.